CPTG: Rotation-Aware Effective-Radius Validation of JWST-SUSPENSE 129982

Carter L Glass Jr

Abstract

Curvature Polarization Transport Gravity, or CPTG, is usually tested against rotation-supported galaxies through circular-velocity structure, while compact quiescent systems can also be tested through effective-radius kinematic reductions. JWST-SUSPENSE galaxy 129982 requires a rotation-aware reduction because its observed stellar kinematics show both a large intrinsic dispersion and strong ordered rotation. This article gives a focused effective-radius CPTG validation of 129982 using the published JWST-SUSPENSE structural and kinematic parameters. The calculation derives the local pressure scale from the intrinsic dispersion and effective radius, computes the stellar baryonic field from the Chabrier-IMF stellar mass, solves the reduced polarization response at the effective radius, and compares the implied rotation-plus-dispersion support with the observed value. The result is a reduced effective-radius validation, not a full radial Jeans reconstruction. The central CPTG velocity estimate falls inside the published observational uncertainty for the measured effective-radius rotation speed.

1 Motivation

Rotation-supported galaxies allow the radial gravitational field to be inferred from circular velocity profiles. Dispersion-supported galaxies require a Jeans-style pressure reduction. JWST-SUSPENSE 129982 occupies the intermediate but physically important case: it is a massive, high-redshift, quiescent galaxy with substantial ordered rotation and substantial random stellar motion.

The JWST-SUSPENSE analysis reports the intrinsic effective-radius rotation speed

$$V_{R_e} = 334^{+52}_{-46}\,\mathrm{km\,s^{-1}}, \tag{1}$$

and the intrinsic dispersion parameter

$$\sigma_0 = 213^{+11}_{-2}\,\mathrm{km\,s^{-1}}. \tag{2}$$

The ratio

$$\frac{V_{R_e}}{\sigma_0} = 1.57^{+0.26}_{-0.23} \tag{3}$$

shows that a pressure-only Jeans reduction would undercount the observed support. The appropriate reduced CPTG test is therefore rotation-aware:

$$a_{\rm obs}(R_e) = \frac{\sigma_0^2 + V_{R_e}^2}{R_e}. \tag{4}$$

The purpose of this paper is narrow: determine whether the observed rotation-plus-dispersion support scale of 129982 is consistent with the CPTG [**?**, 7] reduced polarization response at the effective radius.

2 Rotation-Aware Effective-Radius Reduction

For a galaxy with both random stellar motion and ordered rotation, the scalar effective-radius support can be written as a reduced second-moment scale,

$$V_{\rm rms}^2(R_e) = \sigma_0^2 + V_{R_e}^2. \tag{5}$$

The corresponding observed support acceleration is

$$a_{\rm rms}(R_e) = \frac{V_{\rm rms}^2(R_e)}{R_e}. \tag{6}$$

The pressure component alone defines the local pressure anchor

$$a_\sigma = \frac{\sigma_0^2}{R_e}. \tag{7}$$

This anchor is not treated as a universal acceleration constant. It is the local effective-radius scale set by the measured random stellar motion of 129982.

The CPTG validation then asks whether the stellar baryonic field, acted on by the reduced polarization response, supplies the total support required by

$$a_{\rm rms}(R_e). \tag{8}$$

Equivalently, after the CPTG field is computed, the predicted rotation speed is obtained by subtracting the observed pressure component:

$$V_{\mathrm{CPTG}}(R_e) = \sqrt{g_{\mathrm{CPTG}}(R_e) R_e - \sigma_0^2}. \tag{9}$$

This makes 129982 a direct rotation-aware effective-radius validation target rather than a pure pressure-supported Jeans target.

3 Observed Inputs

The validation uses the scalar quantities needed for an effective-radius test. The adopted source values are the JWST-SUSPENSE Table 1 values for galaxy 129982 [1].

Observed quantity	Adopted value	Validation role
Galaxy ID	129982	JWST-SUSPENSE target
Spectroscopic redshift, z_{spec}	1.249	Epoch and distance scale
Effective radius, R_e	$6.48 \pm 1.13\,\mathrm{kpc}$	Effective-radius scale
Sérsic index, n	3.6 ± 0.1	Structural context
Axis ratio, q	0.39 ± 0.01	Projected shape
MSA position-angle offset, $\Delta\mathrm{PA}$	$30 \pm 1^\circ$	Modelling geometry
Rotation speed, V_{R_e}	$334^{+52}_{-46}\,\mathrm{km\,s^{-1}}$	Ordered support
Intrinsic dispersion, σ_0	$213^{+11}_{-2}\,\mathrm{km\,s^{-1}}$	Pressure anchor
V_{R_e}/σ_0	$1.57^{+0.26}_{-0.23}$	Rotation-aware flag
Spin parameter, λ_{R_e}	$0.52^{+0.07}_{-0.06}$	Fast-rotator check
$\log_{10}(M_{\mathrm{dyn}}/M_\odot)$	11.9 ± 0.1	Dynamical comparator
$\log_{10}(M_\star/M_\odot)$	11.22 ± 0.01	Chabrier source mass

Table 1: Observed inputs used in the rotation-aware effective-radius CPTG validation of JWST-SUSPENSE 129982.

The stellar mass corresponds to

$$M_\star = 10^{11.22} M_\odot = 1.6596 \times 10^{11} M_\odot. \tag{10}$$

The published dynamical mass corresponds to

$$M_{\mathrm{dyn}} = 10^{11.9} M_\odot = 7.9433 \times 10^{11} M_\odot. \tag{11}$$

The dynamical-to-stellar mass ratio is therefore

$$\frac{M_{\mathrm{dyn}}}{M_\star} = 4.7863. \tag{12}$$

This large dynamical excess is precisely the regime in which a reduced CPTG source-response test is useful.

4 Local Acceleration Scales

The adopted effective radius is

$$R_e = 6.48\,\mathrm{kpc} = 1.9995 \times 10^{20}\,\mathrm{m}. \tag{13}$$

The local pressure anchor is

$$a_\sigma = \frac{\sigma_0^2}{R_e}. \tag{14}$$

Using

$$\sigma_0 = 213\,\mathrm{km\,s^{-1}}, \tag{15}$$

gives

$$a_\sigma = 2.2690 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{16}$$

The observed rotation-aware support scale is

$$a_{\mathrm{rms}} = \frac{\sigma_0^2 + V_{R_e}^2}{R_e}. \tag{17}$$

Using

$$V_{R_e} = 334\,\mathrm{km\,s^{-1}}, \tag{18}$$

gives

$$a_{\mathrm{rms}} = 7.8481 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{19}$$

The observed acceleration support is therefore not pressure dominated. The pressure-only scale accounts for

$$\frac{a_\sigma}{a_{\mathrm{rms}}} = 0.2891, \tag{20}$$

while the remaining support is carried by ordered rotation.

5 Reduced Polarization Response

The reduced stellar baryonic field at the effective radius is computed from the measured Chabrier-IMF stellar mass:

$$g_\star(R_e) = \frac{GM_\star}{R_e^2}. \tag{21}$$

With

$$M_\star = 1.6596 \times 10^{11} M_\odot, \tag{22}$$

the effective-radius baryonic field is

$$g_\star(R_e) = 5.5090 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{23}$$

The reduced CPTG polarization response is defined by the local source-to-pressure contrast:

$$\mathcal{R}_{\mathrm{PT}}(R_e) = \left[\frac{g_\star(R_e)}{a_\sigma}\right]^{1/2}. \tag{24}$$

Substituting the computed fields gives

$$\mathcal{R}_{\mathrm{PT}}(R_e) = 1.5582. \tag{25}$$

The CPTG effective field is then

$$g_{\mathrm{CPTG}}(R_e) = \mathcal{R}_{\mathrm{PT}}(R_e) g_\star(R_e). \tag{26}$$

The resulting value is

$$g_{\mathrm{CPTG}}(R_e) = 8.5841 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{27}$$

The effective-radius Structural Mode is

$$N_{R_e} = \frac{g_{\mathrm{CPTG}}(R_e)}{a_\sigma}. \tag{28}$$

Substituting the computed field and pressure scale gives

$$N_{R_e} = 3.7832. \tag{29}$$

6 Validation Results

The central CPTG field is compared against the observed rotation-aware support:

$$\frac{g_{\mathrm{CPTG}}(R_e)}{a_{\mathrm{rms}}} = 1.0938. \tag{30}$$

Thus the central CPTG field is 9.38% higher than the observed reduced second-moment support scale.

The more direct validation is the predicted rotation speed after subtracting the observed pressure component:

$$V_{\mathrm{CPTG}}(R_e) = \sqrt{g_{\mathrm{CPTG}}(R_e)R_e - \sigma_0^2}. \tag{31}$$

Substituting the computed field and the observed dispersion gives

$$V_{\mathrm{CPTG}}(R_e) = 355.35\,\mathrm{km\,s^{-1}}. \tag{32}$$

The observed value is

$$V_{R_e} = 334^{+52}_{-46}\,\mathrm{km\,s^{-1}}. \tag{33}$$

The central residual is

$$\Delta V = V_{\mathrm{CPTG}}(R_e) - V_{R_e} = 21.35\,\mathrm{km\,s^{-1}}. \tag{34}$$

Relative to the upper observational uncertainty, this is

$$\frac{21.35}{52} = 0.41. \tag{35}$$

Relative to the lower observational uncertainty, this is

$$\frac{21.35}{46} = 0.46. \tag{36}$$

The predicted CPTG velocity therefore falls inside the published observational interval:

$$288\,\mathrm{km\,s^{-1}} \leq V_{R_e} \leq 386\,\mathrm{km\,s^{-1}}. \tag{37}$$

The principal numerical results are summarized in Table 2.

Quantity	Observed/input value	CPTG or reduced output
Rotation speed, V_{R_e}	$334^{+52}_{-46}\,\mathrm{km\,s^{-1}}$	$355.35\,\mathrm{km\,s^{-1}}$
Intrinsic dispersion, σ_0	$213^{+11}_{-2}\,\mathrm{km\,s^{-1}}$	Pressure anchor
Effective radius, R_e	$6.48 \pm 1.13\,\mathrm{kpc}$	$1.9995 \times 10^{20}\,\mathrm{m}$
Stellar source mass, $M_\star$	$10^{11.22} M_\odot$	$1.6596 \times 10^{11} M_\odot$
Pressure scale, a_σ	From σ_0^2/R_e	$2.2690 \times 10^{-10}\,\mathrm{m\,s^{-2}}$
Observed support, a_{rms}	From $(\sigma_0^2 + V_{R_e}^2)/R_e$	$7.8481 \times 10^{-10}\,\mathrm{m\,s^{-2}}$
Baryonic field, $g_\star(R_e)$	From $GM_\star/R_e^2$	$5.5090 \times 10^{-10}\,\mathrm{m\,s^{-2}}$
CPTG field, $g_{\mathrm{CPTG}}(R_e)$	Reduced response applied	$8.5841 \times 10^{-10}\,\mathrm{m\,s^{-2}}$
Polarization response, $\mathcal{R}_{\mathrm{PT}}(R_e)$	Source-to-pressure contrast	1.5582
Structural mode, N_{R_e}	Field-to-pressure ratio	3.7832

Table 2: Observed values compared with CPTG reduced effective-radius outputs for JWST-SUSPENSE 129982.

The validation result is positive at the effective radius. The CPTG-predicted rotation speed is higher than the central measured value, but the offset is comfortably inside the quoted observational uncertainty. The object is therefore a useful high-redshift rotation-aware CPTG target.

7 Scope and Falsifiability

This validation is intentionally limited. It is an effective-radius calculation based on scalar published observables. It does not determine the full radial mode function

$$N(r) = \frac{g_{\mathrm{CPTG}}(r)}{a_\sigma(r)}. \tag{38}$$

A full radial validation requires the spatial-bin kinematic profiles used in the JWST-SUSPENSE forward modelling, especially

$$V_\star(R), \tag{39}$$

and

$$\sigma_\star(R). \tag{40}$$

With those data, the radial mode function can be reconstructed directly, the pressure and rotation terms can be separated across radius, and projection effects can be tested without relying on a single effective-radius reduction.

The present calculation also does not replace the published dynamical mass model. The JWST-SUSPENSE dynamical mass uses a kinematic model with rotation, dispersion, geometry, and dynamical assumptions appropriate to the survey analysis. The CPTG calculation here is narrower: it tests whether the Chabrier stellar source at the observed effective radius can produce the measured rotation-plus-dispersion support through the reduced CPTG response.

The immediate falsifiable tests are therefore:

$$V_{\mathrm{CPTG}}(R_e) \approx V_{R_e}, \tag{41}$$

$$\frac{g_{\mathrm{CPTG}}(R_e)}{a_{\mathrm{rms}}(R_e)} \approx 1, \tag{42}$$

and, once the spatial-bin data are available,

$$N(r) = \frac{g_{\mathrm{CPTG}}(r)}{a_\sigma(r)}. \tag{43}$$

A future failure would be clear: if a resolved reconstruction shows that the CPTG field cannot reproduce the observed rotation and dispersion profiles using the same stellar mass and geometry, then the present effective-radius success would be only a scalar coincidence.

8 Conclusion

JWST-SUSPENSE 129982 provides a high-redshift rotation-aware effective-radius test of CPTG. The object is not a clean pressure-supported Jeans-only target because

$$\frac{V_{R_e}}{\sigma_0} = 1.57^{+0.26}_{-0.23}. \tag{44}$$

The measured intrinsic dispersion and effective radius give

$$a_\sigma = 2.2690 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{45}$$

The measured Chabrier-IMF stellar mass and effective radius give

$$g_\star(R_e) = 5.5090 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{46}$$

The reduced CPTG polarization response is

$$\mathcal{R}_{\mathrm{PT}}(R_e) = 1.5582. \tag{47}$$

The corresponding CPTG field is

$$g_{\mathrm{CPTG}}(R_e) = 8.5841 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{48}$$

This field implies

$$V_{\mathrm{CPTG}}(R_e) = 355.35\,\mathrm{km\,s^{-1}}. \tag{49}$$

The observed value is

$$V_{R_e} = 334^{+52}_{-46}\,\mathrm{km\,s^{-1}}. \tag{50}$$

The CPTG result falls within the observed interval. The effective-radius Structural Mode is

$$N_{R_e} = 3.7832, \tag{51}$$

This structural value is reported only as an output of the effective-radius calculation. The validation claim rests on the measured kinematic agreement. The next decisive step is a resolved radial reconstruction from the JWST-SUSPENSE spatial-bin kinematics.

References

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